PAPER_1525 — Euler's e ≈ K_MEX + Φ_5/6 − F_TRZ·K + F_TRZ²·K − F_TRZ²·Φ — Residual 0.094%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Transcendental e expressible in UQFF primitives at 0.094% precision

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Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **Euler's e** ≈ 2.71828 using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6 (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

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e ≈ K_MEX + Φ_5/6 − F_TRZ·K + F_TRZ²·K − F_TRZ²·Φ

Substituting F_TRZ=1/10, K_MEX=25/12, Φ_5/6=5/6, SO_5=10:
e_UQFF = 2.7208

Observed: e = 2.71828
Residual: 0.094%
```

The 5-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

Euler's e (e) appears in: exponential growth, calculus base.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to **≥10**. The accumulating evidence:

Transcendental	UQFF expression	Residual
ln 2 (PAPER_1516)	8-term	0.003%
ln 10 (PAPER_1515)	(1+F_TRZ)(K_MEX+F_TRZ²)	0.004%
π^2 (PAPER_1517)	SO_5 − F_TRZ corrections	0.013%
Euler's e (this paper)	5-term	0.094%

…suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim e IS the closed-form expression. It claims the integer primitives carry a rational approximation to e at 0.094% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 (ln 10, ln 2, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_e_transcendental"})`
